

INTRODUCTION

The database is an essential media form for understanding the logics of digital information systems (Owens, 2018). At the base level, raw bit streams (which have a tangible and objective materiality) are collected, rendered, manipulated, altered, and made usable through the nested assemblages of platforms and protocols. In this sense, data can be perceived as a substance—a solid with objective physical existence. But can it be managed in much the same way as a record? (Yeo, 2018).

A digital object is a multiplicity of orders and sequences whose abstractions coexist with the grammar of computation. This information is then accessed through a database logic: We query orders and sequences of data (or data streams) through a variety of tools, but namely relational databases, processors, and datastores. Or, what we like to call platforms! One might wonder . . . To what extent is this data separate from its media?

Data is at the center of many challenges in system design today. Difficult issues need to be figured out, such as scalability, consistency, reliability, efficiency, and maintainability. What might library systems and standards contribute to such a diverse landscape?

Today, I'll be looking at one particular organization called The Syllabus, examining how they harvest, process, store, and curate data for their clients. Particularly, I am interested in generating a vocabulary that describes the database lifecycle and that efficiently indexicalizes all data entries with search key values in a desired range.

Yet, The Syllabus's lack of standardisation in nomenclature across a diversity of sources with differing data qualities produces hard-to-reconcile data structures. What might the conventions and standards of organizing systems in library science contribute to relational data schemas?

SOURCE TYPE FACET CLASSIFICATION (STFC): RECORD-KEEPING AS INFORMATION ARCHITECTURE

How can LIS professionals contribute to the findability of data, given the availability of many competing services in the information ecology?

Does the management of records differ from the management of information? Is there a wider information landscape that record keeping has seeped into, something we might call information architecture? Or is this a distinct practice with altogether different implications?

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THE STFC SYSTEM

01: Content Type
(what type of information object?)

Article, Book, Journalism, Video, Podcast, Document, Unspecified

02: Language
(what language is the content?)

English, German, French, Italian, Spanish, Portuguese, multilingual

03: Temporal
(time manifestation?)

Year, Month, Day, Published After, Published Before

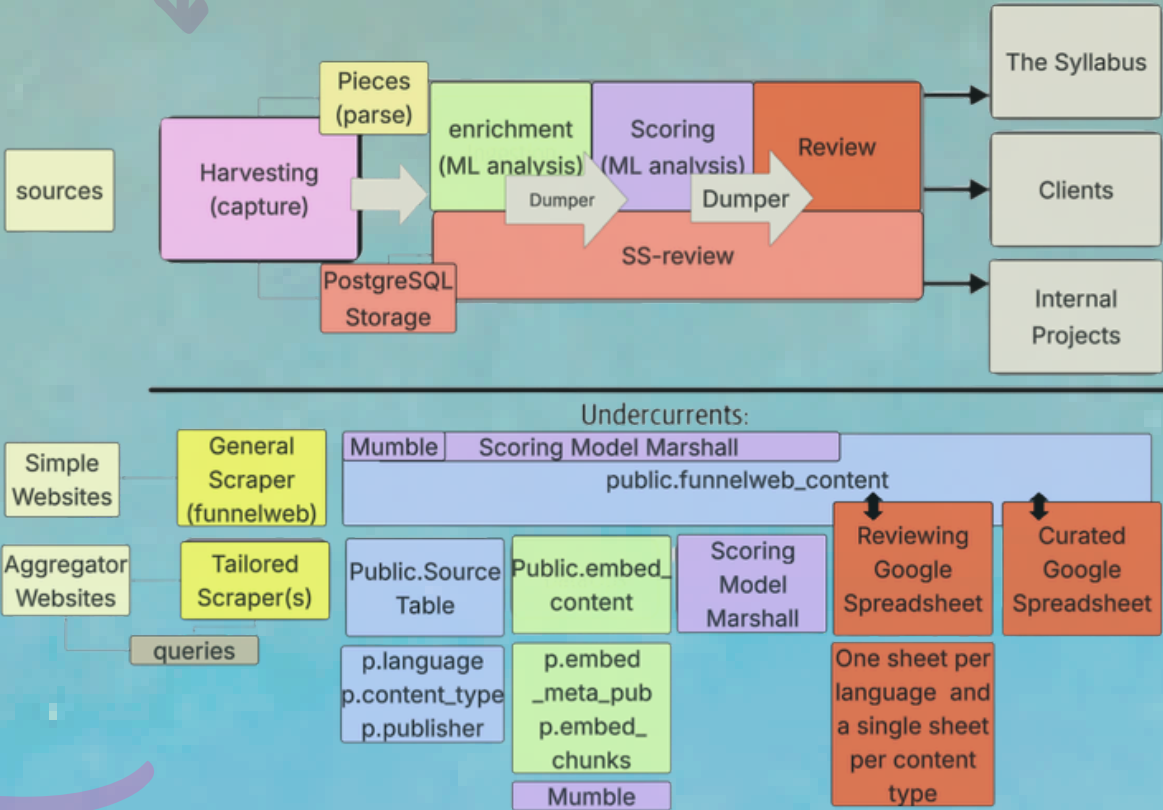
04: Subject/Topic
(what is the content about?)

Section, Module, Unit, Query

05: Domain
(where manifestation?)

Domain, Publisher

LIFECYCLE



FINDINGS & PITFALLS

We have an organizing faceted system that accounts for the transitivity of a particular source across time and space inspired by S.R. Ranganathan's fundamental categories: Personality, Matter, Energy, Space, and Time (Ranganathan, 1967). For the system to work effectively, data must first be transformed into a standardized structural schema for reuse and recall. Yet, the information's materiality, its current organization and configuration, is held back by the underlying architecture of the source system. Data is not necessarily tangible, or queryable. The native multiplicity of sequences and orders essential to the logics of the database still exists of course, but the creation of meaningful orders is being held back? Why?

Acquisition and Description: Challenges associated with data processing; lack of criteria for producing, collecting, and ingesting heterogeneous and diverse data.

Preparation, Assurance, Analysis: The absence of an appropriate method (incomplete normalization) for accommodating and validating a high variety of data

Discovery and Access: Data Quality Issues: language field inconsistencies, malformed author arrays, ad-hoc field names leading to missing interoperability layer and therefore, no standardized query language optimized for indexing.

METADATA AS ARCHITECTURE

The descriptive schema below is designed to facilitate consistent representation across the stages of the pipeline. Through the re-engineering of tables into categories, an environment emerges. All data is local, inseparable from its source system.

